

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: be462587 • Generated: 15:34:08.089Z

Natural Query:	Map surface water bodies and wetlands using NDWI.
Classified Task:	SPECTRAL_INDEX
Calibrated Confidence:	94% (Calibrated via Radiometric Bands)
Specialist Models:	Spectral-Index-Analyzer-v1
Total Latency:	526.7 ms

GROUNDING AI INTERPRETATION

Normalized Difference Water Index (NDWI) delineated hydrological features across the target area (mean NDWI: 0.1841, range: [-0.0811 to 0.9588]). The analysis delineates 38.09% open surface water bodies (e.g., coastal lagoon and marine perimeter) and 43.55% transitional wetland/mudflat. Hydrological classification: Mixed Coastal Lagoon & Wetland.
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INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
isro_sdsc_optical.tif	OPTICAL	512x512	3	EPSG:4326

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (ms)
1	Query Ingestion	Received natural language query: 'Map surface water bodies and wetlands using NDWI.' with 1 uploaded image(s).	0.0
2	Input Validation	Verified 1 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: isro_sdsc_optical.tif	Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326	46.4
4	Modality Identification	Detected sensor modalities: ['OPTICAL'].	0.0
5	Query Classification → SPECTRAL_INDEX	Single image inquiry with spectral / radiometric index mapping intent -> Routed to Spectral Index Specialist (NDVI/NDWI/NDBI).	0.0
6	Model Dispatch	Dispatched task to specialist: 'Spectral-Index-Analyzer-v1' on device 'cpu'.	0.0
7	Specialist Execution Completed	Inference finalized in 479.9ms. Evidence artifacts compiled.	479.9
8	Result Fusion & Calibration	Derived grounded response with calibrated confidence 94% (Calibrated via Radiometric Bands).	0.0